SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1596

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**

Question Count: 4

Duration : 60 Minutes

Code of Conduct

I Pragadeesh Nalan S V(192321161) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pragadeesh Nalan S V

Experiments Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks-1

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Aim:

To create a cloud-based Library Book Reservation System using a Cloud Service Provider to demonstrate the Software as a Service (SaaS) model.

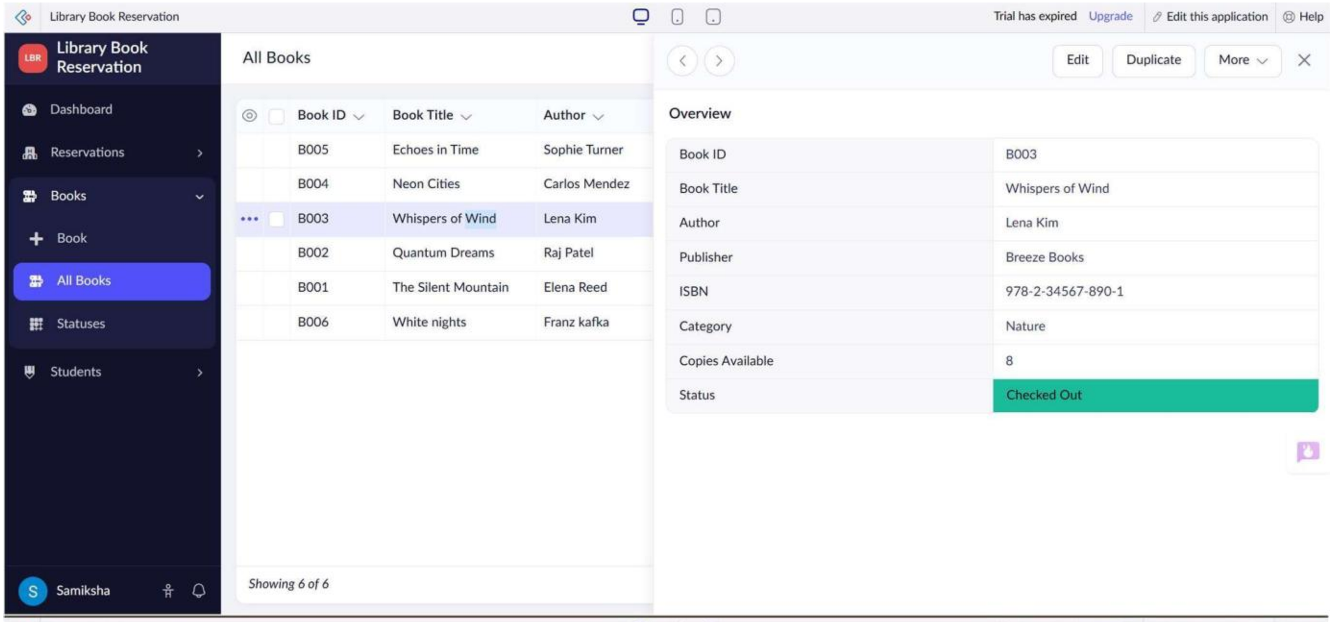
Procedure

1. Open the cloud application development platform.
2. Create forms for managing library books.
3. Add fields such as Book Title, Author, Publisher, Year, Edition, Number of Copies, Rack Number, and Status.
4. Create reservation and return modules.
5. Save the data in the cloud database.
6. Test the application by reserving and returning books.

Result:

A cloud-based Library Book Reservation System was successfully developed and tested. The application effectively manages book information and reservation status using the SaaS model.

Screenshots:


Library Book Reservation
Trial has expired Upgrade Edit this application Help
Library Book Reservation
Dashboard
Reservations
Books
Book
All Books
Statuses
Students
Samiksha
All Books
Book ID Book Title Author
B005 Echoes in Time Sophie Turner
B004 Neon Cities Carlos Mendez
B003 Whispers of Wind Lena Kim
B002 Quantum Dreams Raj Patel
B001 The Silent Mountain Elena Reed
B006 White nights Franz kafka
Showing 6 of 6
Edit Duplicate More
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Book Title Whispers of Wind
Author Lena Kim
Publisher Breeze Books
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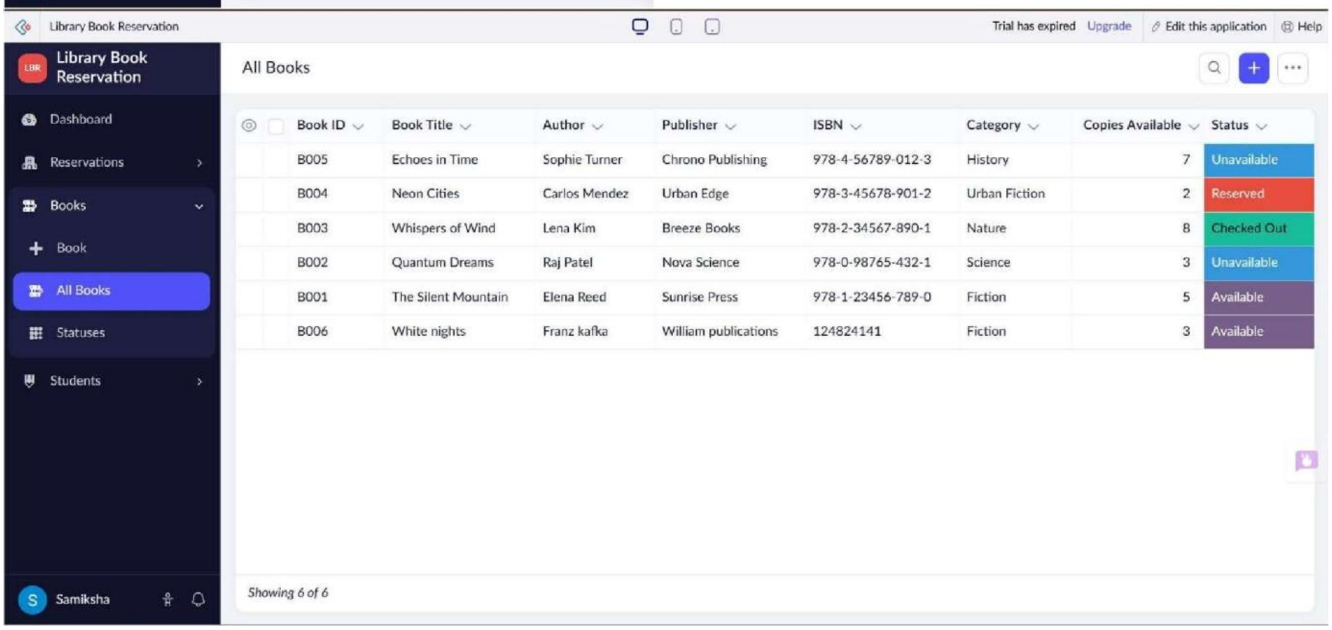
Category Nature
Copies Available 8
Status Checked Out


Library Book Reservation
Trial has expired Upgrade Edit this application Help
Library Book Reservation
Dashboard
Reservations
Books
Book
All Books
Statuses
Students
Samiksha
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B005 Echoes in Time Sophie Turner Chrono Publishing 978-4-56789-012-3 History 7 Unavailable
B004 Neon Cities Carlos Mendez Urban Edge 978-3-45678-901-2 Urban Fiction 2 Reserved
B003 Whispers of Wind Lena Kim Breeze Books 978-2-34567-890-1 Nature 8 Checked Out
B002 Quantum Dreams Raj Patel Nova Science 978-0-98765-432-1 Science 3 Unavailable
B001 The Silent Mountain Elena Reed Sunrise Press 978-1-23456-789-0 Fiction 5 Available
B006 White nights Franz kafka William publications 124824141 Fiction 3 Available
Showing 6 of 6

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Aim

To develop a cloud-based Payroll Processing System using a Cloud Service Provider to demonstrate Software as a Service (SaaS).

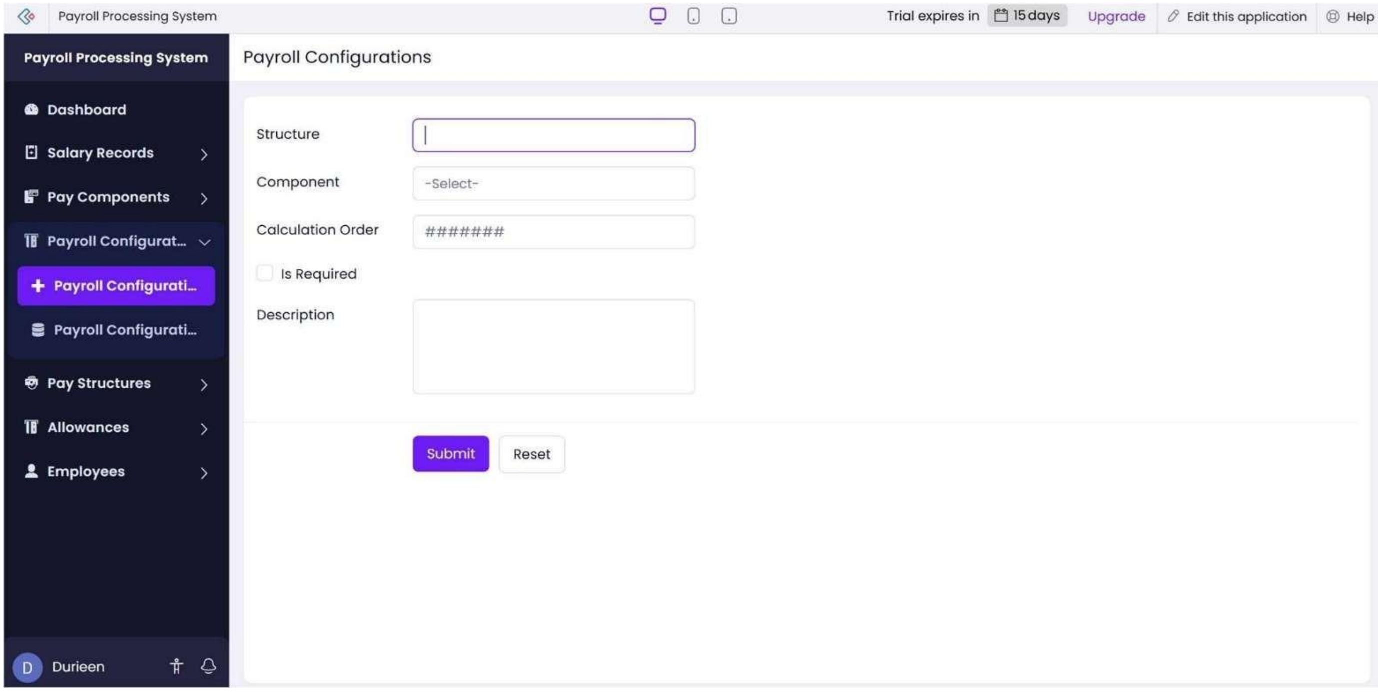
Procedure

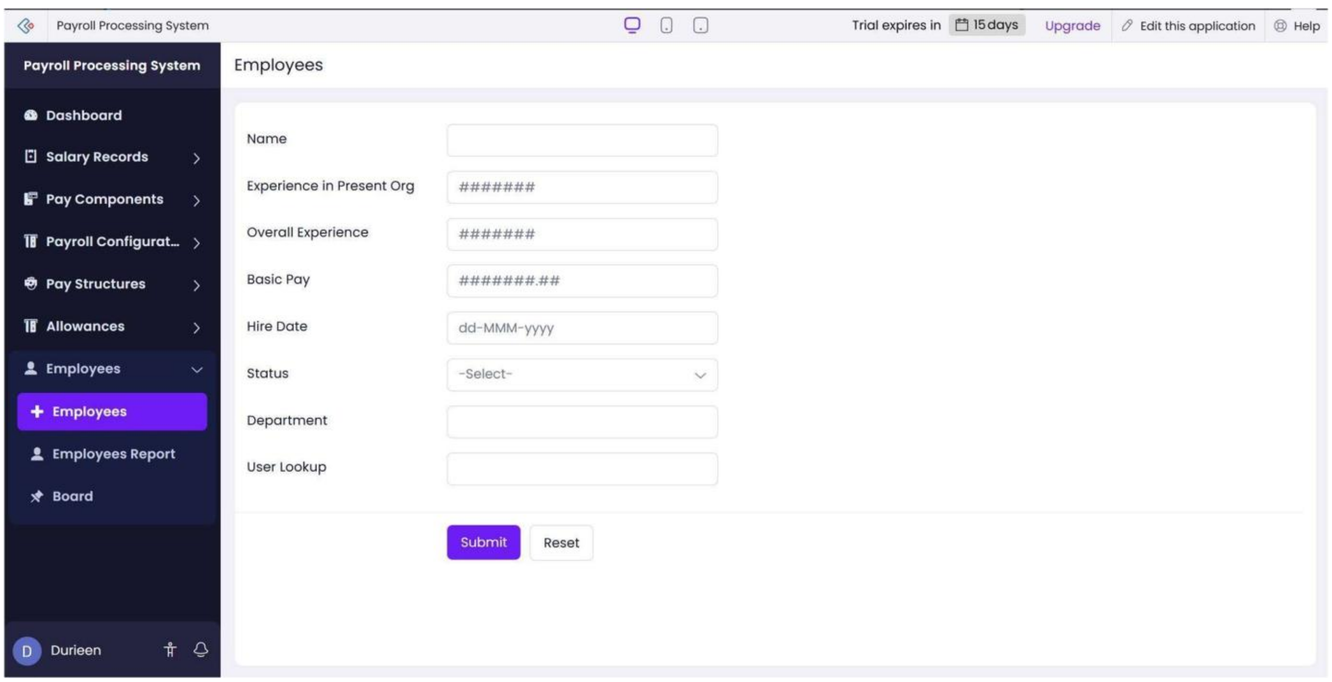
1. Create employee and payroll forms.
2. Add fields such as Employee Name, Experience, Basic Pay, HRA, DA, TA, Gross Pay, and Net Pay.
3. Configure salary calculation formulas.
4. Store payroll records in the cloud database.
5. Generate payroll details and verify the calculations.

Result

A cloud-based Payroll Processing System was successfully implemented. The system accurately calculated employee salaries and maintained payroll records using the SaaS model.

Screenshots:





Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim

To demonstrate virtualization by installing a Type-2 Hypervisor and creating a virtual machine with 1 CPU, 2 GB RAM, and 15 GB storage.

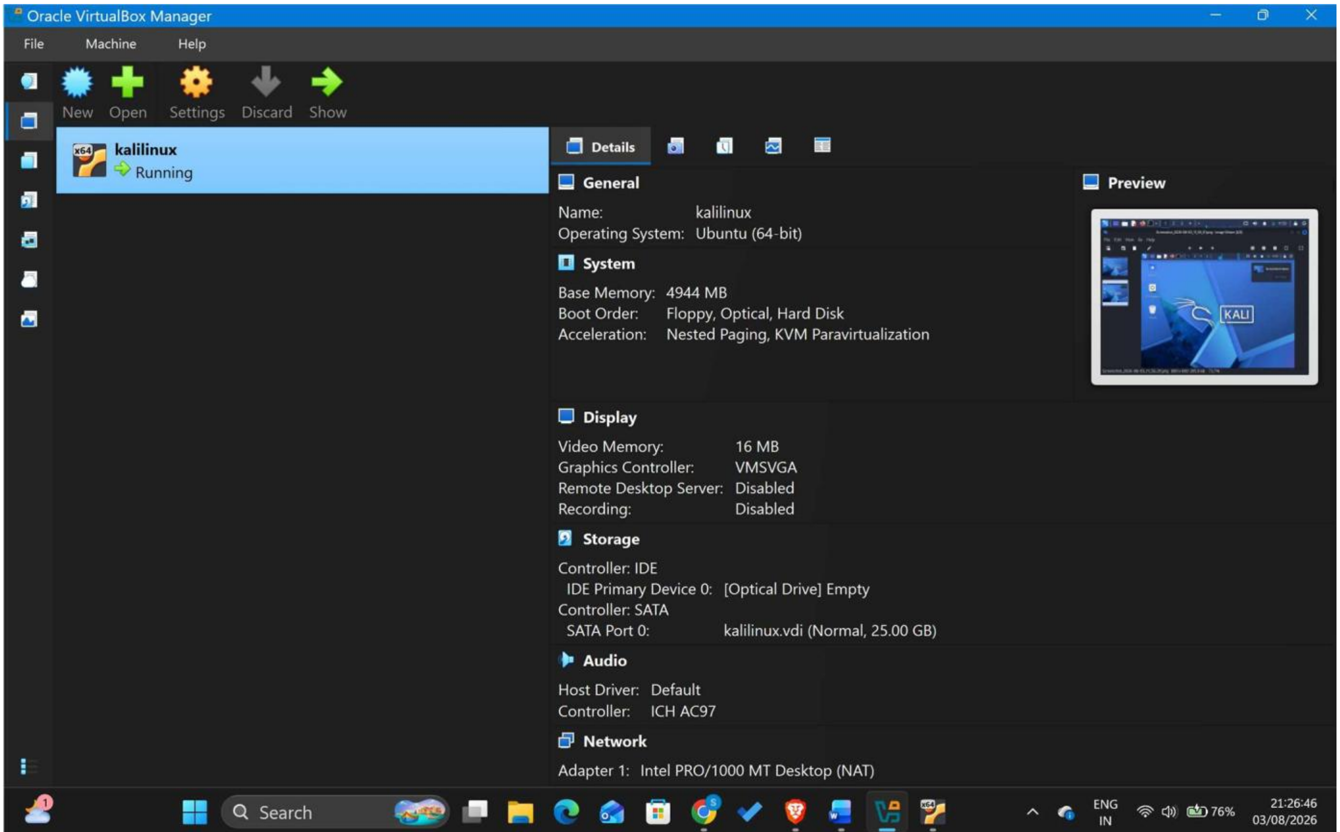
Procedure

1. Install a Type-2 Hypervisor such as Oracle VirtualBox.
2. Create a new virtual machine.
3. Allocate 1 CPU, 2 GB RAM, and a 15 GB virtual hard disk.
4. Install Windows or Linux as the guest operating system.
5. Start the virtual machine and verify its operation.

Result

A virtual machine was successfully created and configured using a Type-2 Hypervisor. The guest operating system booted and functioned correctly

Screenshots:



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Aim

To demonstrate virtualization by cloning an existing virtual machine and testing the cloned virtual machine.

Procedure

1. Open the Type-2 Hypervisor.
2. Select the existing virtual machine.
3. Use the Clone option to create a duplicate virtual machine.
4. Start the cloned virtual machine.
5. Verify that the cloned virtual machine works correctly.

Result

The virtual machine was successfully cloned and tested. The cloned virtual machine operated correctly with the same configuration as the original VM

Screenshots: